University of California, Santa Cruz
Board of Studies in Computer Engineering

CMPE 293: Introduction to Feedback Control System Design
General Information and Syllabus

Spring 2004
T-Th 2:00-3:45 PM
Class: Porter College Room 241

Instructor: Gabriel Hugh Elkaim
E-mail: elkaim@soe.ucsc.edu
Office: Baskin Engineering, 353B
Hours: T-Th 4:00-5:30 PM
Phone: (831) 459-3054 (Office)
(831) 421-9098 (Home: 10AM to 9PM please)
(415) 377-5723 (Cell: 10AM to 9PM please)

TAs and Tutors: TBD.

Required Textbooks (in the library on reserve):

> "Feedback Control of Dynamic Systems (4th Edition) by Franklin, Powell, and Emami, Prentice-Hall, NJ, 2002.

Introduction: Design of linear feedback control systems for command-following error, disturbance rejection, stability, and dynamic response specifications. Root-locus and frequency response design (Bode) techniques. Nyquist stability criterion. Design of dynamic compensators. Examples are drawn from electrical, mechanical, and aeronautical applications. Computer aided design with MATLAB.

Prerequisites: EE103 or equivalent (Differential Equations, Circuits, Physics)

Homework: 1 set per week, due in my office, Thursday at 5 PM.

Website: www.soe.ucsc.edu/classes/cmpe293/Winter04

Check this site often as this is where the homework assignments, lecture notes, homework and test solutions, and lecture videos are posted. You are expected to read the material on the website.

Videos: As an experiment in teaching technology, the instructor will be capturing both the audio and screen from the course in real-time. This will later be posted to the website. While every attempt will be made to capture the classes, as this is a new technology, there are no guarantees. Also, while watching the video should be a great way to review course material, if you are not in class you cannot ask questions and/or clarifications. Further, if too few students show up for lectures, the instructor may restrict access to class videos.

Blog: www.robotica-exotica.com

This is the professor's personal blog, having to do with robotics, control systems, and just technology in general. This is just given as a reference.

Syllabus:

Chapter 1: Intro and History
Chapter 2: Dynamic Models
Chapter 3: Dynamic Response
- Convolution
- Laplace Transforms
- Block Diagrams
- Control Specifications

Chapter 4: Basic Properties of Feedback
Chapter 5: Root Locus Design
Chapter 6: Frequency Response Design
- Frequency Response
- Nyquist
- Bode
- Compensation

Chapter 7: State-Space Design

Grading: the course grade is the weighted average of the following criteria:

Homeworks	20%	
Midterm	30%	04-May-2004
Final Exam	50%	09-June-2004, 8 – 11 AM

If you have any disability-related needs, be sure to contact the Disability Resource Center well in advance of any expected need.

MATLAB: this course will rely heavily on the use of MATLAB for homework, and as a tool to understanding control system behavior. MATLAB is available on the cluster computers in Baskin Engineering, as well as on the Solaris machines. You can also

purchase a copy of the student version of MATLAB directly from the Mathworks (and also from the bookstore). The control system toolbox is also available from the Mathworks.

Academic Honesty

Academic honesty is a requirement for the course. All assignments must be your own independent work; this includes homework and exams.

What is cheating? It is presenting work that is not yours as your own. Using unauthorized reference materials, reproducing the solution off of an old homework are all unacceptable. You can, and are encouraged to, discuss and strategize with your colleagues on homeworks, but your work should be your own. Copying is NEVER acceptable.

If a student is caught cheating in this class this will result in an immediate failure in the class. It will be reported to your college and your department. DO NOT CHEAT; it is not worth it.

Acknowledgements

I would like to acknowledge the help of Prof. Ed Carryer of Stanford University in pioneering this video capture technology, and helping me to set this up. I would also like to thank Prof. Steve Rock of Stanford University for the course material, syllabus, and general inspiration and help in setting up this class.